IMT School PhD Programme in

"Economics, Analytics and Decision Sciences"

Research Statement

A nation's economic development is based on scientific economic research, which also forms the basis of evidence-based policymaking. In order to make a meaningful and substantial contribution in this area, one must develop both applied theoretical and quantitative skills, which are essential for creating workable solutions to actual economic issues.

As a young, aspirational economist, my career and professional aspirations are focused on scientific research since I firmly believe in it. In collaboration with the German Federal Ministry (KfW), the Kiel Institute for the World Economy offered me the chance to enrol in a master's program in applied quantitative economics at the University of Rwanda (UR). This program enhanced the fundamental research and quantitative skills needed for evidence-based policymaking in Rwanda. I recently received a professional diploma in data science from the African Institute of Mathematical Science's (AIMS) Data Science Fellowship Program.

Working as a macroeconomic and climate modelling economist at Rwanda's Ministry of Finance and Economic Planning[1] (MINECOFIN) and as a research fellow at research think tanks like the International Centre for Insect Physiology and Ecology (icipe), GIZ-Deutsche Gesellschaft für Internationale Zusammenarbeit, and Economic Policy Research Network of Rwanda piqued my interest in combining economic modelling with quantitative research for evidence-based policy making. My master's thesis examining the effects of push-pull technology on food security for small-scale

[1] My involvement in the operationalization of the Computable General Equilibrium (CGE) Model for Rwanda and the Quarterly Projection Model (QPM), a near-term forecasting instrument

farmers in Rwanda, which received funding from the Centre for Insect Physiology and Ecology (ICIPE), sparked my interest in applied quantitative impact analysis in agriculture and plant health.

My research interests are focused on macroeconomics, climate analysis, and policy impact evaluation using quantitative research tools such as econometrics, economic modelling, and data science (big data, machine learning). I am investigating the application of connecting economic knowledge with data science development in practical research, driven by my current duties and academic advancements. Additionally, I'm interested in learning more about how macroeconomic modeling techniques, such as the computable general equilibrium (CGE) model for climate analysis, can be extended. I hope to use these methods, for example, to assess the effects of fiscal policy. "Quantifying the sectoral effects of fiscal subsidies—particularly in agriculture, including fertilizer subsidies—on the economy and households' welfare using a CGE modelling approach."

Therefore, pursuing PhD studies in fields like data science and quantitative economics would not only enhance my prior interests and experience but also provide me with the empirical, technical, and quantitative skill set required to fulfill my research career goals and contribute to evidence-based policy decisions in Rwanda and the surrounding area. I'm sure that enrolling in a PhD program in Economics, Analytics, and Decision Sciences (EADS) will help me realize my goal of connecting economic models with data science (AI and machine learning algorithms) to create a contemporary hybrid form of economic analysis that will enable me to take advantage of today's changing technological landscape.